Dennis G. Pezan

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Education

United States Military Academy at West Point — West Point, NY
B.S., Operations Research (Finance Specialization), Minor in Artificial Intelligence — Expected May 2029

- CQPA: 3.571 | Dean's List, Fall 2025 and Spring 2026
- **Stokes Writing Fellowship** – endowed fellowship funding research travel, expert mentorship, and publication of original research
- **Sea to Shining Sea Fellowship** – engages cadets with senior leaders in national security, government, finance, and technology

Research Experience

Contested Logistics / Network Interdiction Research – D/Math, USMA — West Point, NY
Undergraduate Researcher (Advisor: LTC Matthew Webb) — Jul 2026 – Present

- Built a 57-node, 79-edge model of Taiwan's crude oil imports with seven costed adversary actions, verified by independent max-flow
- Score each adversary plan by the share of Taiwan's crude demand still delivered; the RL agent must beat random (about 90%) and greedy (70%) plans

Reinforcement Learning for Contested Resupply – TDAC, U.S. Army — Fort Lee, VA
Research Intern, Modeling & Analysis Division — Jul 2026

- Redesigned the reward function for a PPO reinforcement-learning agent in the Army's Logistics Battle Command simulation
- Diagnosed four reward flaws; the retrained agent more than tripled requests filled and beat a priority-aware baseline over 101 seeds
- Presented findings to Branch Chief LTC Ryan Miller and, by call, to TDAC's AI/RL researchers at the Naval Postgraduate School

Tree-Based Methods in Machine Learning – D/Math, USMA — West Point, NY
Independent Study (MA289), Advisor: LTC Patrick Kuiper — Oct 2025 – May 2026

- Designed and completed an independent study of decision trees, bagging, and random forests; presented the findings to officers

Projects

Tree-Based Methods Research Site — dennis-pezan.com/ma289.html

- Compared decision trees, bagging, and random forests on Titanic data (scikit-learn) by tree correlation and overfitting

CFT Packing List – Web Application — dennis-pezan.com/cft.html

- JavaScript web tool that helps cadets organize and track gear for Cadet Field Training; deployed on Vercel from GitHub

Mounger Writing Center Website Redesign — dennis-pezan.com/mwc-redesign.html

- Independent redesign concept in Next.js, TypeScript, and Tailwind CSS with a custom WebGL ink shader

Professional Development

Technology and Innovation AIAD – Department of Social Sciences, USMA — California and Colorado
Summer AIAD on defense technology — Jun 2026

- Discussed how defense tech is built, bought, and fielded with Palantir, Gordian Knot, OpenAI, Apple, Meta, and 4th Infantry Division

Leadership & Activities

Center for Enhanced Performance — West Point, NY
Cadet-in-Charge, MA103 Tutoring — Aug 2026 – Present

- Lead the full tutor staff for MA103 Data-Driven Modeling (1,010 cadets); built practice exams with answer keys

Mounger Writing Center – *Writing Consultant* — Aug 2026 – Present

- Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

Center for Leader Development in STEM (CLDSTEM) – *Member* — Oct 2025 – Present

- Bring hands-on STEM projects to 800+ students across the Chicagoland and Greater Los Angeles areas

Cadet Ambassador Program – *Cadet Ambassador* — Jan 2026 – Present

- Represent West Point in outreach to prospective cadets

McDonald's — Plainfield, IL
Department Manager — Sep 2022 – Jul 2024

- Promoted at 16, the region's youngest department manager; scheduled 80+ staff, ran $20K+ weekly orders

Skills

Python (PyTorch, scikit-learn), SQL, Java, JavaScript/TypeScript | Reinforcement learning, network optimization | Spanish (fluent)